



SARLiquid: Through-package Liquid Leakage Detection Based on mmWave SAR Imaging

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Background

➤ Why it matters?

Liquid leakage not only cause economic loss and microbial growth. More seriously, for high-concentration alcohol and other volatile liquids, leaks can pose serious safety risks (such as fire or the release of harmful vapors).



Fire

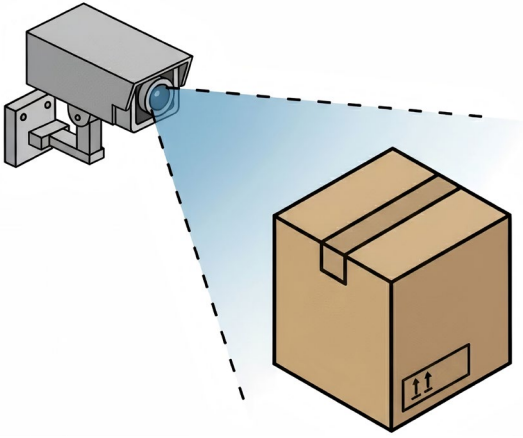


Explosion



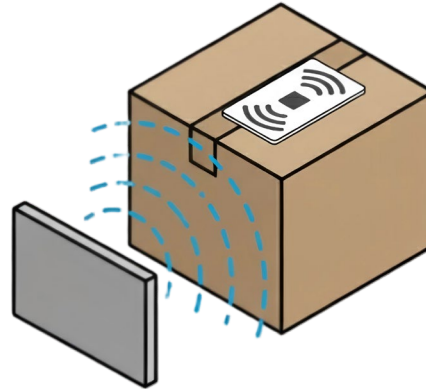
Poisoning

Limitations of Current Methods



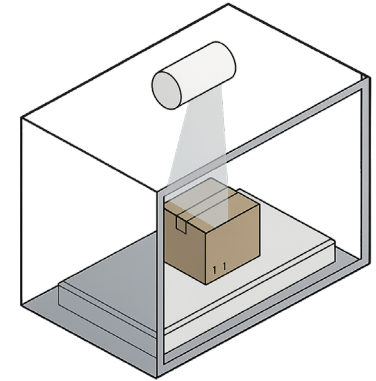
Vision-based methods

Need Line-of-Sight (LoS),
cannot penetrate packaging.



RFID-based methods

Require direct contact with the
leaked liquid.



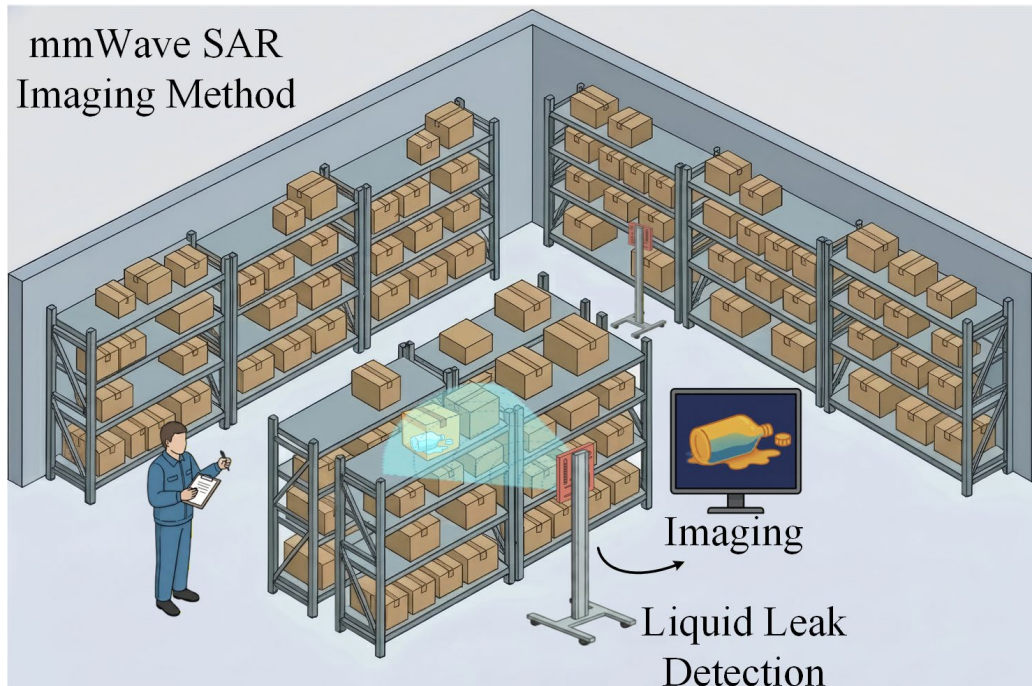
X-Ray-based methods

Effective but poses ionizing
radiation risks.

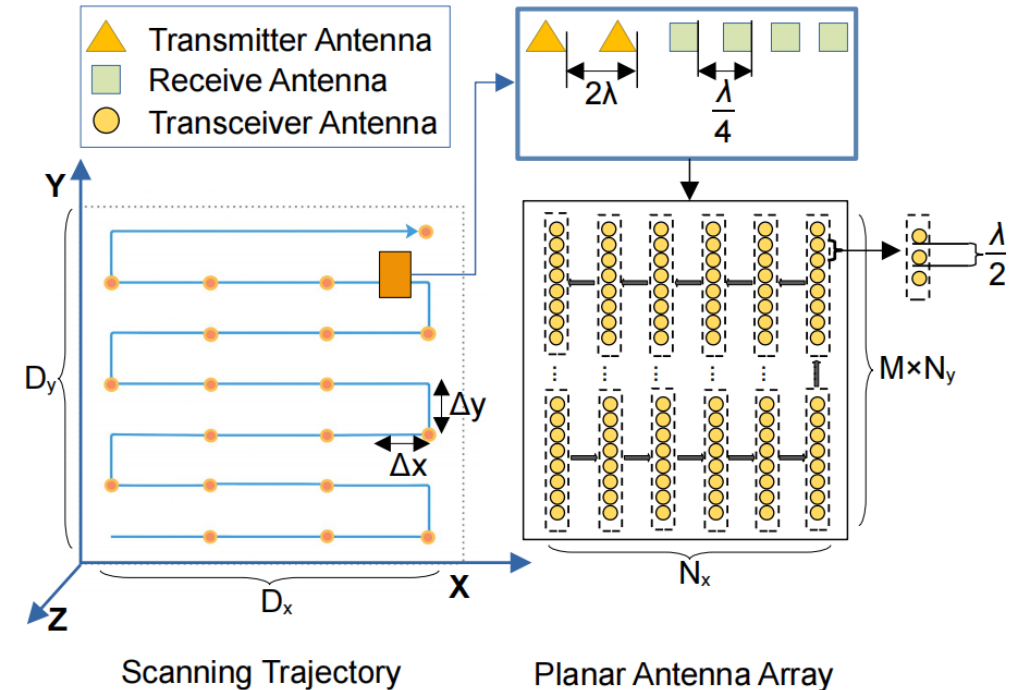
Our Method

What is SAR?

Synthetic aperture radar(SAR) synthesizes a much larger effective antenna aperture by coherently combining echoes collected as the sensing device moves, achieving high resolution sensing with a physically small antenna.

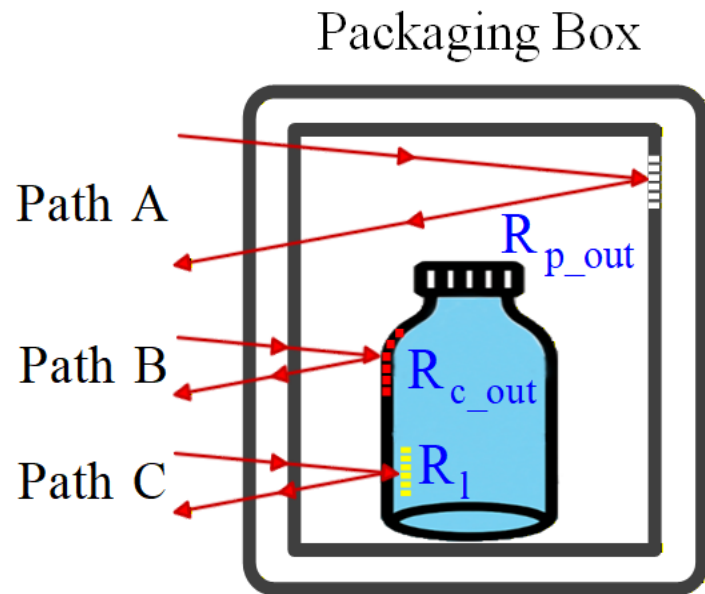


(a)

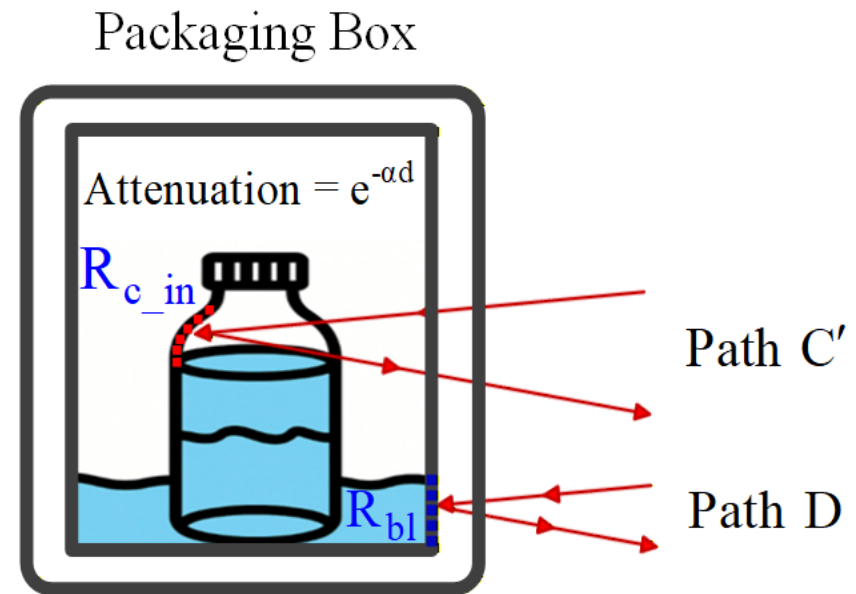


(b)

Insight



There are three paths for the non-leakage scenario

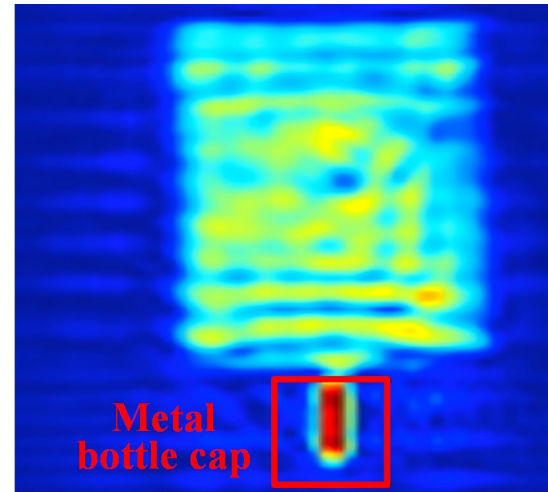
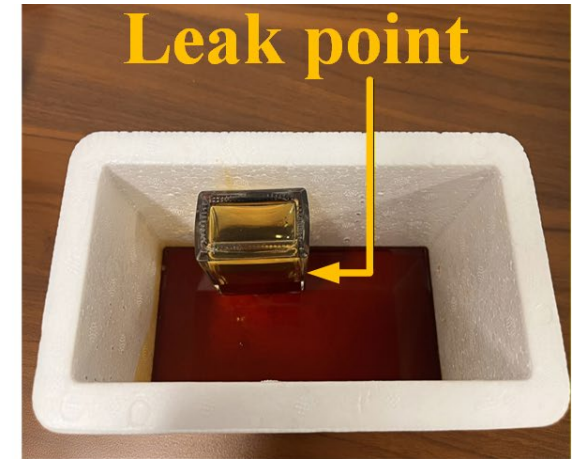


Under the leakage condition, the liquid level in the container decreases, and the leaked liquid spreads to the bottom of the package.

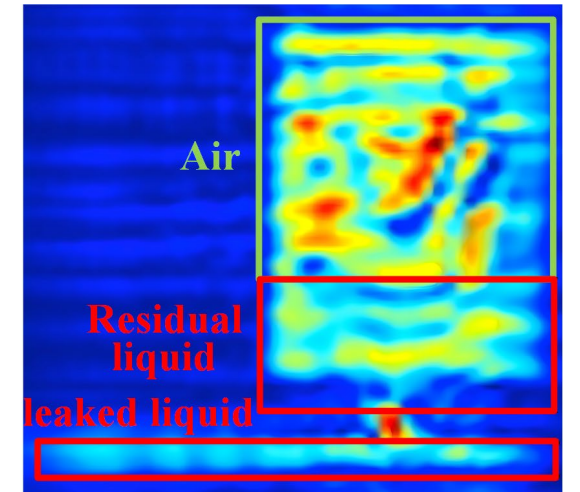
Image reconstruction

We adopt a standard FMCW SAR reconstruction pipeline and reconstruct the **complex SAR image** as

$$f(x, y) = FFT_{2D}^{-1} \left[FFT_{2D} \left[S(x', y') \right] \cdot e^{-jz \sqrt{4k^2 - k_x^2 - k_y^2}} \right]$$



(a) Non-leakage

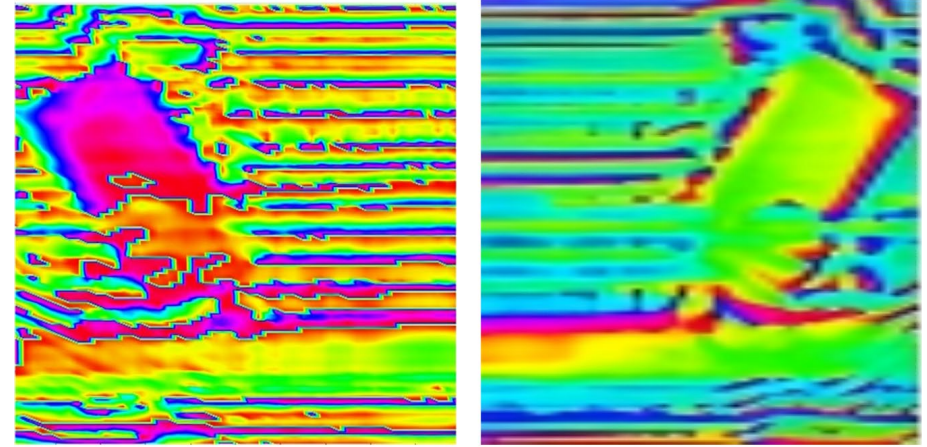


(b) Leakage

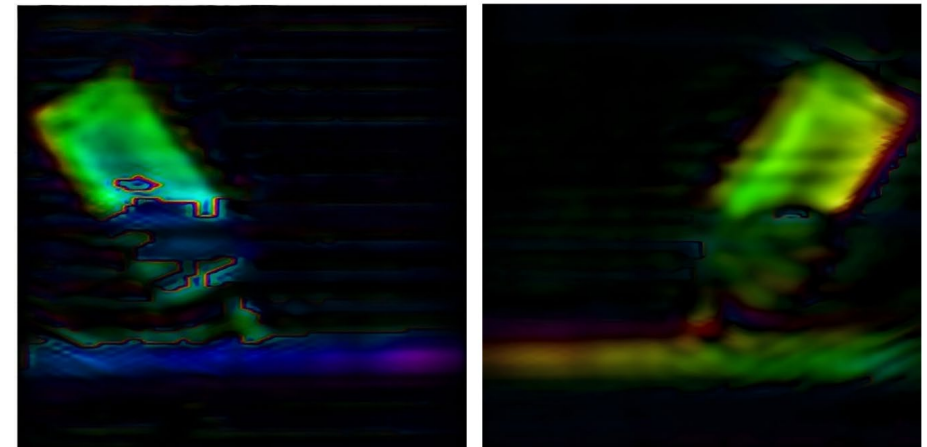
Phase Calibration

We develop a dedicated phase calibration process for restoring phase continuity in phase unwrapping.

- $\Phi_{GF}(x, y) = GF(\Phi(x, y), \Phi(x, y); \sigma_s, \sigma_r)$
- $\Phi_n(u, v) = \Phi_f(u, v) \cdot M(u, v)$
- $\Phi_{uw}(x, y) = SolvePoissonByDCT(\rho)$

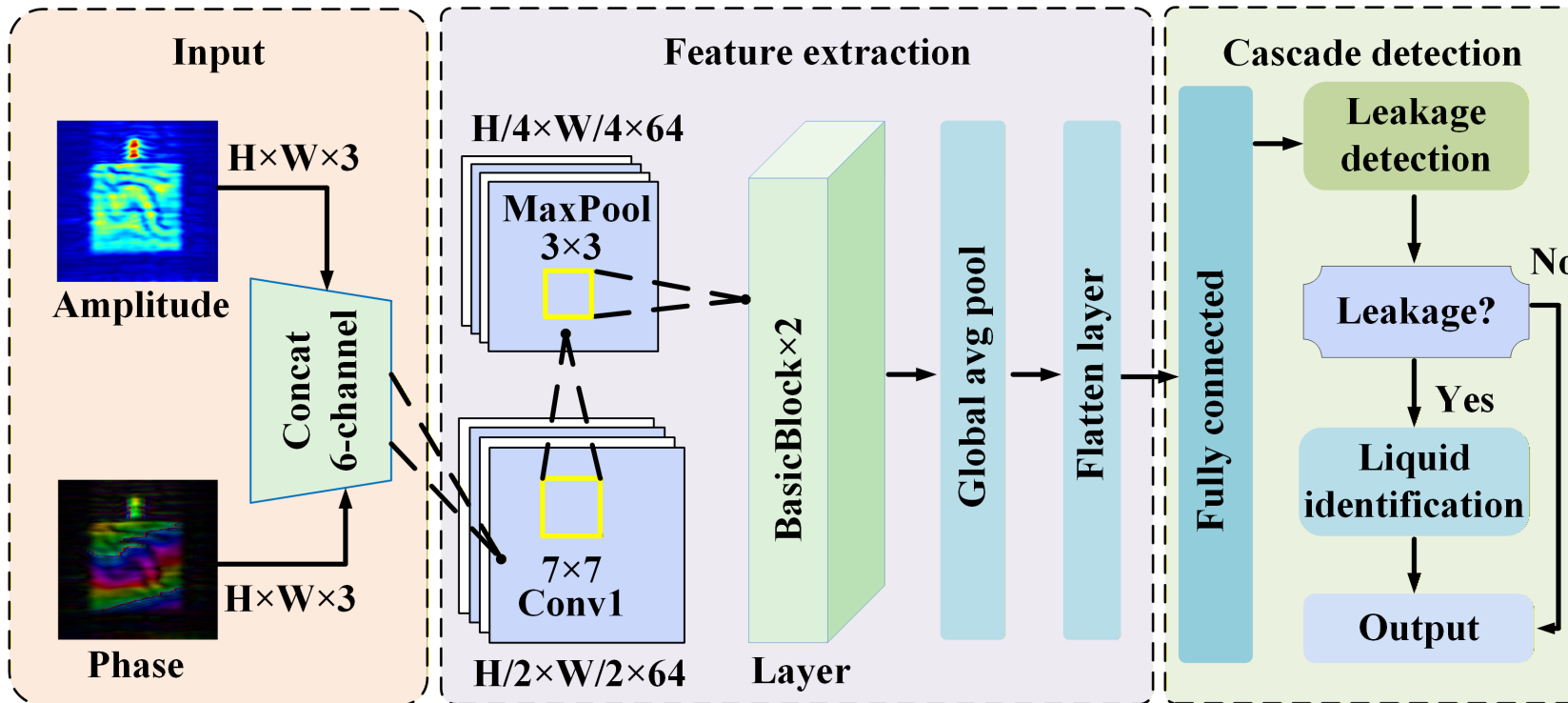


(a) Phase image without calibration



(b) Phase image with calibration

LF-Net



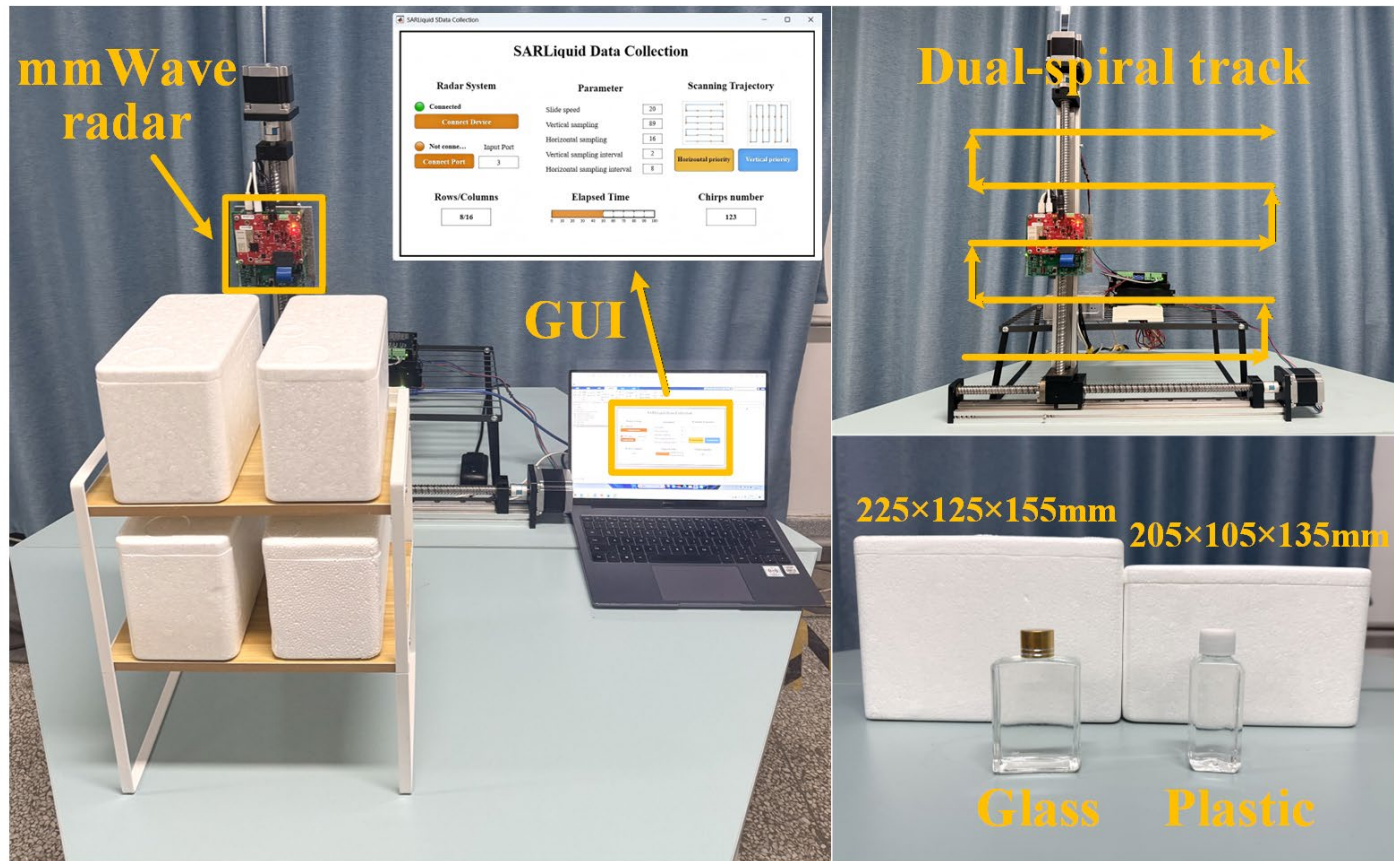
Two cascaded branches

- 1) *Leakage detection branch*
- 2) *Liquid identification branch*

A cascaded loss Function:

$$L_{leak} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=0}^1 y_{i,c}^{leak} \log(\hat{y}_{i,c}^{leak})$$

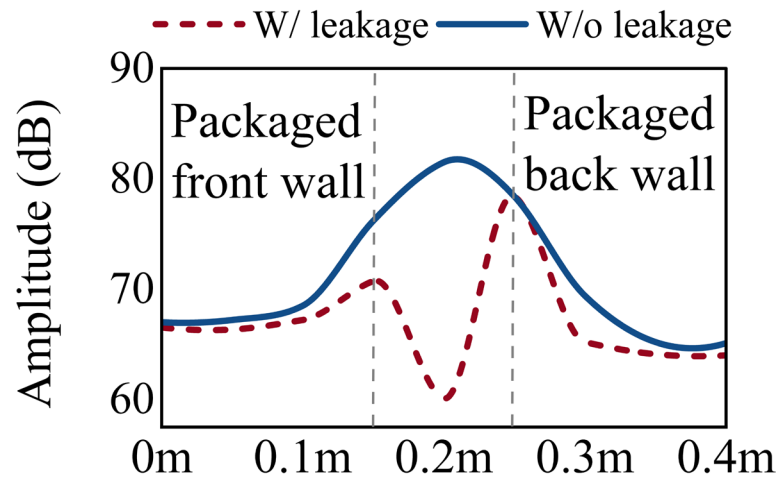
Implementation



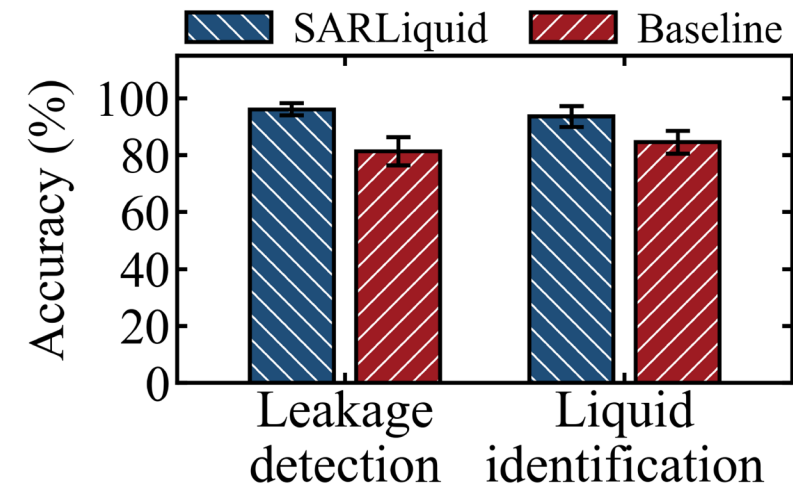
Equipment List

- **mmWave Radar:** TI IWR1843BOOST
- **Mechanical scanning trajectory:** Dual-spiral track
- **Dataset:** 2040 samples

Evaluation



(a) baseline

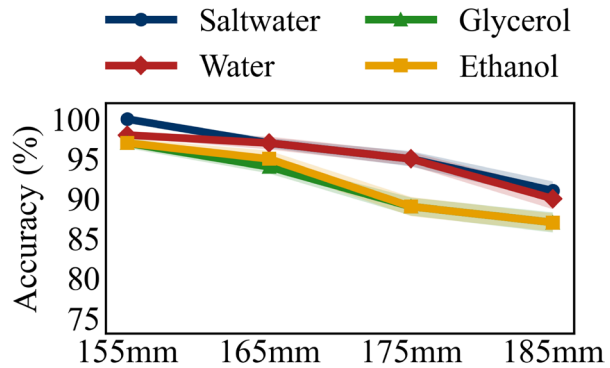


(b) Overall performance comparison

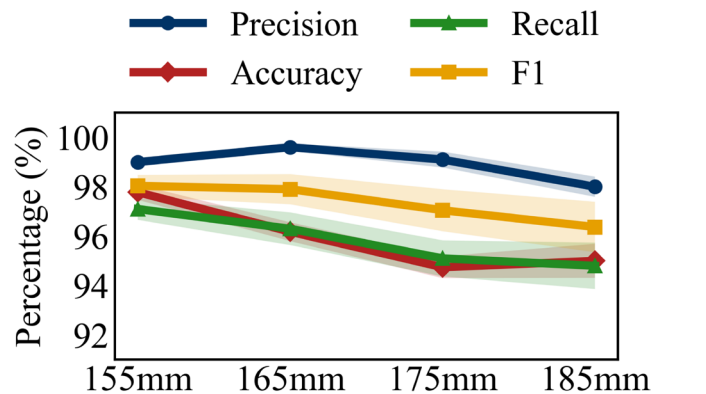
we implement a baseline that analyzes mmWave signals without SAR imaging.

Evaluation

Sensing distance

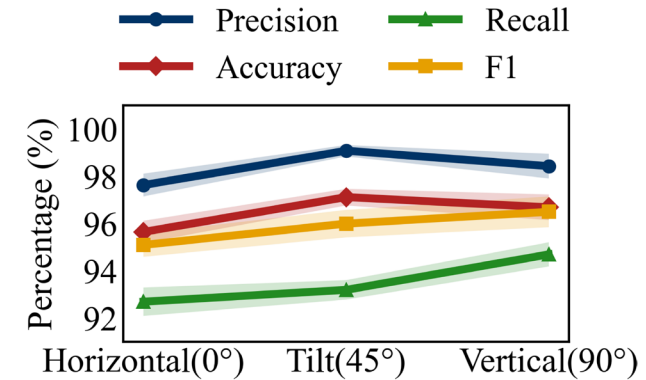


(a) Performance of leak detection

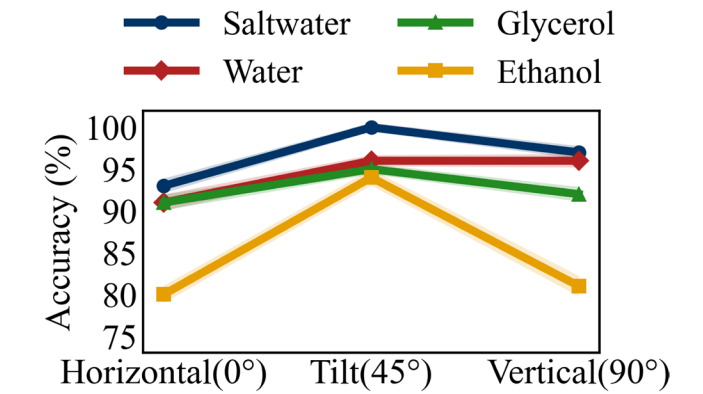


(b) Performance of liquid identification

Container deployment



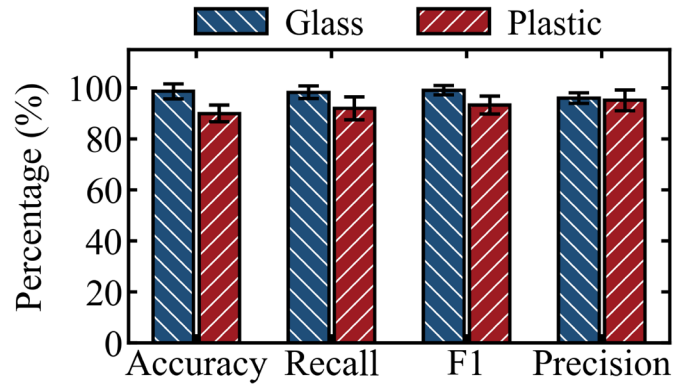
(a) Performance of leak detection



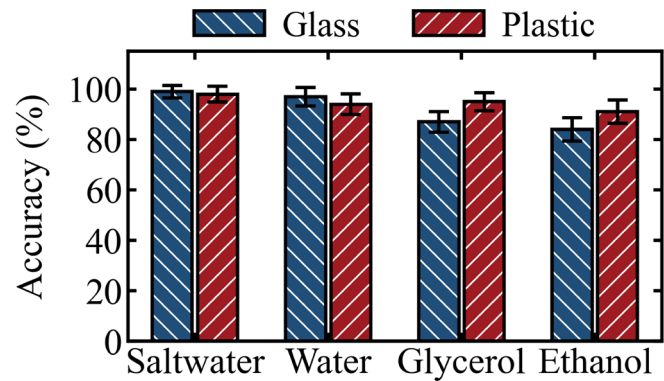
(b) Performance of liquid identification

Evaluation

Container material

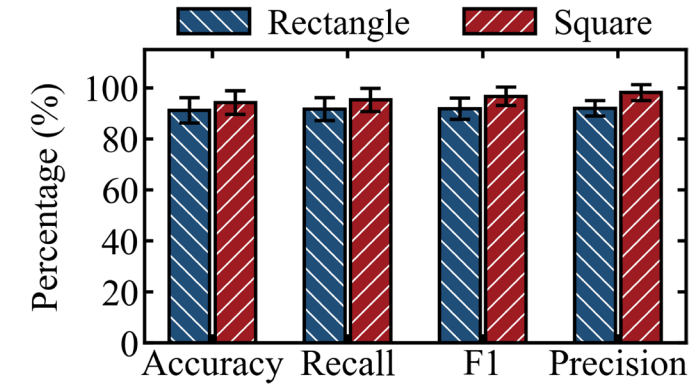


(a) Performance of leak detection

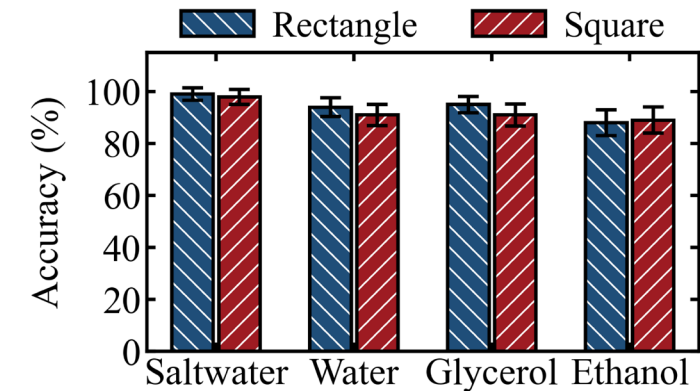


(b) Performance of liquid identification

Container shape



(a) Performance of leak detection



(b) Performance of liquid identification



Conclusion

- We propose SARLiquid, a **contactless** technique for through package **liquid leakage detection** based on mmWave radar.
- We disclose the **propagation model** for mmWave signals in packaged scenarios and establish the relationship between liquid leakage and its features in mmWave images.
- We design the **imaging process**, dedicated **phase correction algorithms**, and a joint **network model** for leakage detection and liquid identification.
- Our experiment results show that SARLiquid achieves **accurate** and **robust** estimation across diverse environmental conditions.

Thank You!



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